# Branch Location Strategies for Thai Banks and Access to Financial Services

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March 24, 2021

**Abstract**

We argue that the effect of financial crises on bank branch location choices provides an unexplored channel by which crises affect access to credit. We study the location strategies of Thai banks and estimate a dynamic structural model of oligopolistic location choice, allowing for complementarity in payoffs for bank branches in nearby locations, as well as competitive effects between rival banks. We predict the evolution of bank branch locations under the counterfactual scenario of no financial crisis in 1997. We find that there would have been 14.3% more branches and 9.1% more markets with at least one branch after ten years had the crisis not occurred.

# 1 Introduction

Countries that suffer a financial crisis often see the real economy seriously impacted. Naturally, a major concern during a financial crisis is whether households and firms have access to credit. A standard way to evaluate when access to credit normalizes is to look at when aggregate measures of economic activity, such as GDP growth and interest rates, return to pre-crisis levels. However, we identify a new channel by which financial crises impact access to credit that can be much longer-lived than would be suggested by aggregate measures: local access to a physical bank branch. This channel is also particularly important for developing countries.

There is a wide literature documenting the effect of physical bank branch proximity on access to banking services (Nguyen, 2019; Agarwal and Hauswald, 2010; Ergungor, 2010; Petkov, 2019; Assuncao et al., 2020; Alem and Townsend, 2014; Ho and Ishii, 2011; Petersen and Rajan, 2002; Degryse and Ongena, 2005; Crawford et al., 2018). One reason for this is transportation costs, but another reason is that bank branch proximity can lower the information collection costs required to assess the viability of loans. Developing countries also typically have incomplete branching networks with significant gaps in coverage, especially in rural areas. Because financial crises particularly affect the functioning of banks, a financial crisis can cause banks to restrict the expansion of their branch networks, or even reduce the size of their networks. To the extent that banks fail to replace these branches, even after the economy recovers, the effects of the crises can be long-lived, and can negatively impact local communities long after aggregate measures of growth suggest the effects of the crisis

are over.

We explore this issue in Thailand, which suffered a major financial crisis in 1997. Aggregate measures of economic activity recovered relatively quickly. For instance, GDP growth and unemployment recovered within two to three years. Interest rates declined shortly after the crisis under IMF support and stayed below pre-crisis levels ever since. However, we show that the crisis had a major impact on branching behavior by commercial banks in Thailand. Entry of new branches fell dramatically for several years after the crisis and, for essentially the first time in Thailand's history, we observe the closure of bank branches. As we document, even when entry rates recovered, entry was not always in the places that saw exit. Several communities that experienced exit still have not seen new entry by the end of our sample period. We also argue that the lack of liquidity forced banks to close branches in rural areas that would have otherwise been profitable in the long run.

Studying the impact of the crisis on branch locations is challenging because there are many large banks in Thailand that have many branches throughout the country. These banks may interact in complex ways that are difficult to describe with simple statistics. To provide a more concrete measure of the impact of the crisis on branch locations, we specify a dynamic structural model of the bank branch location problem and estimate the model using data on branch locations obtained from the Bank of Thailand.

In our model, banks choose whether or not to enter in a large number of heterogeneous locations around Thailand. Branch profits depend on the number of nearby branches of their own bank and of rival banks. Profits also

depend on local demand, which we measure using the intensity of nighttime light surrounding branch locations. We assume that branches beyond a distance threshold do not affect a branch's profits, which allows us to cluster branching locations into separate markets, similar to Zheng (2016). Banks form expectations about the shocks that rivals will realize in the future and account for the benefit of preempting rivals in their branching strategies. We assume the financial crisis in 1997 arrives unexpectedly for the banks and we allow their strategies to change in response to the crisis. Because our environment is nonstationary, we assume the model has a finite horizon and estimate the model using a full solution approach with backward induction, similar to Igami (2017, 2018). We control for persistent market-level unobserved heterogeneity by partitioning markets into five different types. We follow an approach similar to Collard-Wexler (2013) and Lin (2015) to group markets. In our framework, the equilibrium choice probabilities are allowed to differ across market types.

In both our reduced-form and structural results, we find that banks prefer to locate their branches in areas with higher local demand and away from their own and rival branches. Although the financial crisis of 1997 lowered our nighttime luminosity measure of local demand in most markets, we also include an additional indicator for the crisis in the banks' profit functions. This indicator captures the change in demand that is not captured by the observed changes in nighttime luminosity, as well as how the liquidity crisis affected the banks' branching strategies. We estimate a large negative value for this crisis indicator which makes banks less likely to open new branches

and more likely to close existing branches.

Our structural model is able to accurately predict the aggregate expansion and contraction patterns of the branching network. We use the estimated structural model to simulate several counterfactual experiments. First, we simulate the bank branch locations that would have been chosen if there had never been a crisis in 1997, quantifying the effect of the crisis on the bank branch network. We do this by removing the crisis indicator in banks' profit functions, as well as the fall in nighttime luminosity. We find that the expansion of the network followed a path similar to the pre-crisis period and did not experience a contraction. Ten years after the crisis, there would have been 14.3% more branches and 9.1% more markets served by at least one bank in the absence of the crisis. The average distance to the nearest branch would also have fallen by 23.6%. We use the estimated effect of distance on access to commercial loans found by Ji et al. (2021) using Thai data to evaluate the effect of the crisis on financial access. Using their estimates with our change in distance, access to loans would have increased by 10.5 percentage points in the absence of the crisis.

In a second counterfactual experiment, we consider the effect of a branch support subsidy during the post-crisis period on banks' branching strategies. The support we consider is equal to half of the estimated losses from our post-crisis indicator in the branch's profit function. This counterfactual can also be interpreted as easing the liquidity shortages faced by the banks during the crisis. We find that although such a support is not sufficient to allow banks to continue along their pre-crash trend following the crisis, it does succeed

in preventing the proportion of markets served by at least one branch from falling below pre-crisis levels. Ten years following the crisis, there are 6.0% more branches and 4.6% more markets served by at least one bank with the supports. These results motivate the rationale of such supports, as the crisis caused banks to close branches that may have been long-term profitable. The cost of entry is a large multiple of a branch's typical annual profits. If a branch's losses and liquidity issues are large enough to force the bank to close a branch during the crisis, the large entry cost may preclude the bank from reopening a branch in that location after the crisis, even if that branch would have been profitable in the long run. However, supporting a branch for the duration of the crisis would mean that location could continue to be served following the crisis.

# 2 Background and Data

## 2.1 The 1997 Financial Crisis

From 1985-1996, Thailand had the highest rate of economic growth in the world. During this time, it maintained a low inflation rate, low unemployment and a stable exchange rate. The exchange rate was tied to a basket of dominant world currencies, with a high weight on the US dollar. Thailand's high growth and stability therefore made it very attractive to foreign investors. However, a number of shocks made it difficult to maintain a fixed exchange rate. The real estate boom resulted in supply eventually exceeding demand, causing the number of vacancies to increase and borrowers to default on their loans. The

FIGURE 1: Thailand's GDP growth and unemployment rate. Data source: World Bank

US also raised interest rates, which diverted investment away from Southeast Asia. The country then had a current account deficit for several years and the central bank's foreign reserves were insufficient to maintain a fixed exchange rate. In May 1997, with an imminent move towards a flexible exchange rate regime, there were speculative attacks from currency traders. The speculative attacks became a self-fulfilling prophecy when Thailand eventually let their currency float in July 1997. The Thai Baht immediately experienced an enormous devaluation and the economy went into crisis.

Soon after, the IMF stepped in to help stabilize the economy. Figure 1 shows the GDP growth and unemployment rate during this period. GDP growth was negative for only two years and then returned to a healthy growth rate of around 5%. Even during the height of the crisis, unemployment only reached 3.5% and by 2002 it had fallen to 1.5%. Therefore we might conclude that Thailand recovered from the crisis within a few years. As we will see, however, the slowdown in branch openings and the closures of existing bank branches continued on until 2003-2004, and the effects of the closures were

long-lived in some areas.

## 2.2 The Thai Banking Industry

Our bank branch data are from the Bank of Thailand and contain information on all bank branches operating in Thailand from 1927-2010. For each bank branch we observe the open date, close date (if any) and GPS coordinates. Our data do not include branches from the Bangkok Metropolitan province or the neighboring province Samut Prakan, which make up the Greater Bangkok Area.

There are 18 different commercial banks in our data. The commercial banks combined had 3,730 bank branches across the country in 2010. In our analysis, we focus on the 4 largest commercial banks: Bangkok Bank, Kasikorn Bank, Krung Thai Bank and Siam Commercial Bank. These four banks constitute over two-thirds of the total number of commercial branches in our last period of data and each have significantly more branches than all of the smaller banks. Krung Thai Bank is a state-owned bank, but all four banks are publicly-traded companies.

Government banks also operate in Thailand. There are two main government banks with a total of 1,928 branches at the end of 2010. These are the Government Savings Bank (GSB) and the Bank for Agriculture and Agricultural Cooperatives (BAAC), which have 499 branches and 1,429 branches respectively. The BAAC do not tend to locate their branches in urban areas and their motives are less likely to be profit-oriented (see Assuncao et al. (2020)). The GSB, on the other hand, does locate its branches in more urban

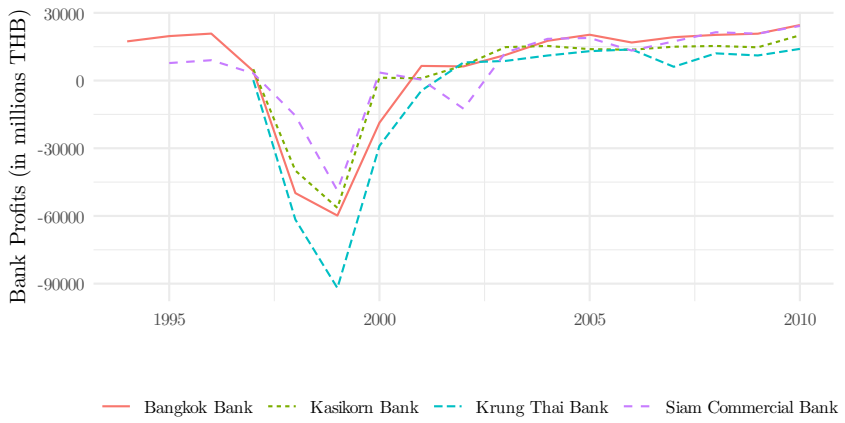



FIGURE 2: Net profit per bank from the banks' annual reports (in millions of Thai Baht).

areas, with the primary aim of mobilizing savings. There is very little presence of foreign banks outside of the Greater Bangkok Area.

The 1997 financial crisis had a large effect on the commercial banks operating in Thailand. Using information from the four largest banks' annual reports, we show each bank's net profits over time in Figure 2. We can see that each of the four largest banks were severely affected by the crisis and showed similar patterns. Profits remained negative for several years before recovering.

In the years following the 1997 crisis, banks slowed the expansion of their branch networks. For the first time in Thailand's history, there were bank branch closures, mainly in rural areas. In many locations where bank branches closed, it was many years before the bank branches were replaced, if they were replaced at all. Figure 3 shows an example area in northern Thailand that was badly affected by the crisis. The red points denote bank branch locations, the gray lines show the road network, and the colors in the heatmap show the distance to the nearest bank branch. On the eve of the crisis in 1996, the area

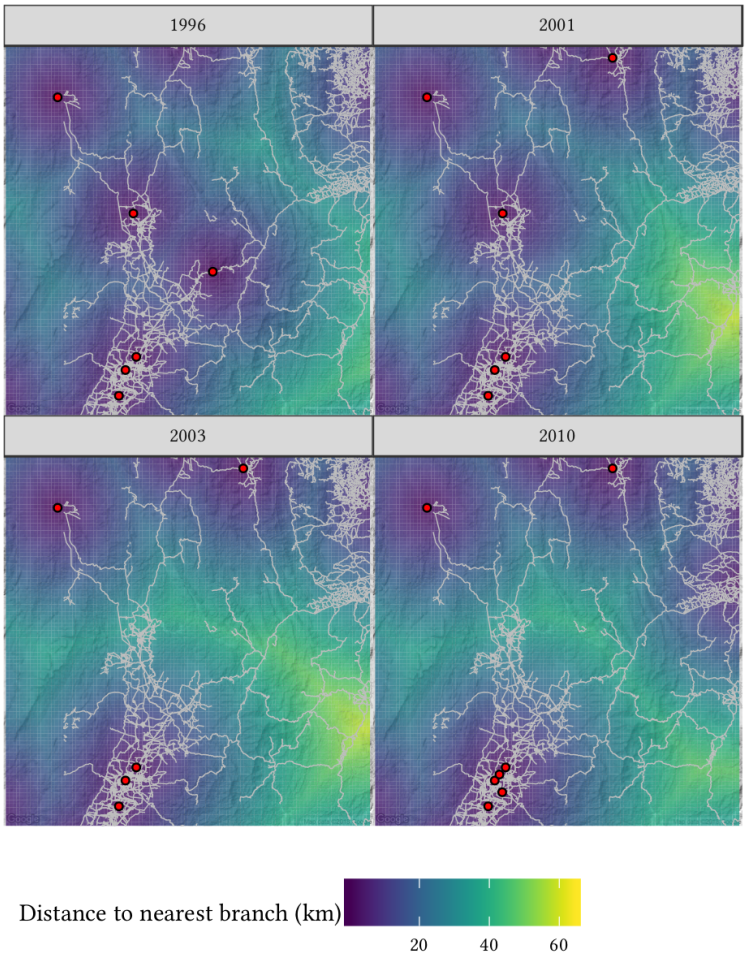



FIGURE 3: Distance to nearest commercial branch in the Phrae changwat following the financial crisis, 1996, 2001, 2003 and 2010.

in the center of the map was reasonably well-served by branches with most locations being within 20km of a branch. Following the crisis, one branch closed in 2001 and another closed in 2003. Even by the end of our sample period in 2010, these locations that saw their branches close did not see a new one reopen, leaving them very far from the nearest branch. Although the number of bank branches on aggregate recovered some years after the crisis, there were a large number of areas which became underserved for many years.

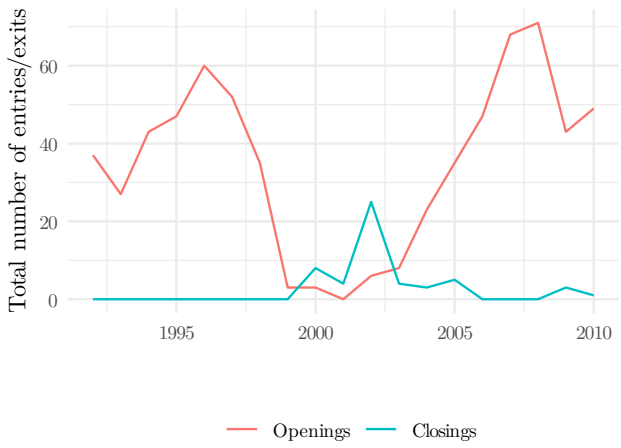



FIGURE 4: Number of openings and closings by year for the largest four banks.

The crisis had an immediate effect on the opening of new branches, but banks did not begin closing existing branches until 2-3 years later. This can be seen in Figure 4 which plots the total number of branch openings and closings per year by the four largest banks in our sample. We can see a reduction in branch openings starting immediately after the crisis in 1998 and lasting until 2004. Branch closings began in 2000 and peaked in 2002. According to the 1996 financial report of Siam Commercial Bank, they had anticipated to open 30 branches in 1997, but only opened 22 and did not close any branches. In 1999, they stated they "slowed domestic branch expansion and reassessed the potential of existing branches." Only in their 2001 report do they state they had "implemented a rationalization program" that "resulted in merging and closing down of branches."

# 3 Model

## 3.1 Overview

We now describe our model for how banks make their branch-network expansion decisions. In our model, banks make independent branching decision in each market. A bank's profits in a market depends on local demand, the number of branches they own and the number of rival branches. The financial crisis arrives unexpectedly and has a negative effect on profits. Banks are forward-looking and strategic in their their branching decisions. Banks take into account the responses of their rivals to their actions and how they may affect their future profits.

## 3.2 Model Setup

There are $F$ commercial banks who can simultaneously choose to open and close branches in $M$ different markets in each period $t$. If bank $f$ has $n_{fmt}$ active branches in market $m$ at time $t$, it earns profits equal to:

$$\pi_{fk}\left(\boldsymbol{s}_{mt}, \boldsymbol{\theta}\right) = n_{fmt}\left(\theta_k + \theta_{bf} + \theta_z z_{mt} + \theta_{own}\left(n_{fmt} - 1\right) + \theta_{comp}\sum_{g \neq f} n_{gmt} + \theta_{crisis}\zeta_t\right)$$

Each market $m$ belongs to one of $K$ types, and the term $\theta_k$ in the profit function is allowed to differ by market type, $k = 1, \ldots, K$. The per-branch profit is also allowed to differ by bank and this is captured by $\theta_{bf}$, for $f = 2, \ldots F$, where we make the normalization $\theta_{b1} = 0$ for bank 1. The state variable $z_{mt}$ is an

exogenous market demand shifter that affects profits. We assume $z_{mt}$ follows a Markov process that is independent of the banks' actions. The parameter $\theta_{own}$ measures the agglomoration or cannibalization effect of the bank's own branches. If $\theta_{own} > 0$, then a branch benefits from having another branch of the same bank nearby. If $\theta_{own} < 0$, new branches cannibalize profits from its existing branches. The parameter $\theta_{comp}$ measures the competitive effect of nearby branches of rival banks.

The term $\theta_{crisis}$ measures the effect of the financial crisis on profits that is not captured by changes in $z_{mt}$. This term also captures the effect of the banks' lower liquidity on their payoffs; for example, if the bank cannot borrow to make additional loans. The crisis arrives as a surprise for the banks. Before the crisis occurs, banks believe $\zeta_t = 0$ in all future time periods. After the crisis arrives, $\zeta_t$ follows a deterministic process that is known by the banks. The market state, $\boldsymbol{s}_{mt}$, is the combination of each bank's number of branches, $\{n_{fmt}\}_{f=1}^{F}$, the demand shifter $z_{mt}$ and whether there is a financial crisis, $\zeta_t$. This is summarized by $\boldsymbol{s}_{mt} = \left(\{n_{fmt}\}_{f=1}^{F}, z_{mt}, \zeta_t\right)$.

A bank's profits within a market are assumed to only depend on the local demand shifter and the presence of nearby own and rival branches. Therefore branch profits are independent of any of the banks' actions in other markets. Banks are also assumed to be risk neutral and have no geographic diversification motives such as those described in Aguirregabiria et al. (2016). This simplification means each bank's national branching problem can be solved with independent branch network decisions in each market.

We assume the set of available actions for firm $f$ in market $m$ at time $t$ is to

open one branch, close one branch or maintain the same number of branches. A single bank cannot open or close more than one branch in the same market in the same time period. A bank can also have at most $N$ branches in a market. Denote the firm's action by $a_{fmt} \in \{-1, 0, 1\}$, where $-1$ denotes closing a branch, 0 denotes maintaining the same number of branches and $+1$ denotes opening a branch. The set of available actions for firm $f$ in market $m$ at time $t$, $\mathcal{A}_f(n_{fmt})$, therefore depends on their existing number of branches:

$$\mathcal{A}_f(n_{fmt}) = \begin{cases} \{0, 1\} & \text{if } n_{fmt} = 0 \\ \{-1, 0, 1\} & \text{if } n_{fmt} \in \{1, \ldots, N-1\} \\ \{-1, 0\} & \text{if } n_{fmt} = N \end{cases}$$

Each bank chooses to open or close branches simultaneously within a time period. Choosing to open or close a branch only takes an effect with a one-period lag. We can therefore write the process for a bank's number of branches in a market as $n_{fmt+1} = n_{fmt} + a_{fmt}$. If a bank chooses to open a branch, they incur the cost of entry $\theta_{ec}$. The scrap value from closing a branch is normalized to zero because it would not be separately identified from the entry cost, $\theta_{ec}$, and the constant terms, $\theta_k$. Banks also receive action-specific private information shocks $\boldsymbol{\varepsilon}_{fmt} = \left(\varepsilon^{-1}_{fmt}, \varepsilon^{0}_{fmt}, \varepsilon^{1}_{fmt}\right)$ which affect their payoffs. We assume these payoffs are independent and follow a Type I extreme value distribution.

## 3.3 Equilibrium

Banks are forward-looking and discount future profits with a discount factor $\beta \in (0,1)$. In a terminal period $T$ the market state is fixed and no longer changes. The profits of active branches in a market remain the same forever starting from period $T$. The value function for bank $f$ in market $m$ of type $k$ in the terminal period $T$ is then:

$$\widetilde{V}_{fkT}\left(\boldsymbol{s}_{mT}, \boldsymbol{\theta}, \boldsymbol{\varepsilon}_{fmT}\right) = \frac{\pi_{fk}\left(\boldsymbol{s}_{mT}, \boldsymbol{\theta}\right)}{1-\beta}$$

The value function for bank $f$ in market $m$ of type $k$ for time periods $t < T$ is:

$$\begin{aligned}\widetilde{V}_{fkt}\left(\boldsymbol{s}_{mt}, \boldsymbol{\theta}, \boldsymbol{\varepsilon}_{fmt}\right) = \pi_{fk}\left(\boldsymbol{s}_{mt}, \boldsymbol{\theta}\right) + \max_{a \in \mathcal{A}_f\left(n_{fmt}\right)} \Big\{ & \varepsilon^a_{fmt} - \theta_{ec}\mathbb{1}\left\{a=1\right\} \\ & + \beta \mathbb{E}\left[\widetilde{V}_{fkt+1}\left(\boldsymbol{s}_{mt+1}, \boldsymbol{\theta}, \boldsymbol{\varepsilon}_{fmt+1}\right) \middle| \boldsymbol{s}_{mt}, a_{fmt} = a\right] \Big\}\end{aligned}$$

The bank earns its flow profits in period $t$ and, based on the realization of the private information shock $\boldsymbol{\varepsilon}_{fmt}$, chooses the action that maximizes its expected present discounted value of payoffs. Banks form expectations of the future market state based on the transition probabilities of the Markov process $z_{mt}$ and their beliefs of their rivals' strategies.

As the private information shocks are distributed Type I extreme value, the expected value function before the realization of the private information

shock is given by:

$$V_{fkt}\left(\boldsymbol{s}_{mt},\boldsymbol{\theta}\right)=\pi_{fk}\left(\boldsymbol{s}_{mt},\boldsymbol{\theta}\right)+\quad\log\left(\sum_{a\in\mathcal{A}_f\left(n_{fmt}\right)}\exp\left\{-\theta_{ec}\mathbb{1}\left\{a=1\right\}+\right.\right.$$
$$\left.\left.\beta\mathbb{E}\left[V_{fkt+1}\left(\boldsymbol{s}_{mt+1},\boldsymbol{\theta}\right)\middle|\,\boldsymbol{s}_{mt},a_{fmt}=a\right]\right\}\right)$$

Similarly, the probability that bank $f$ chooses action $a\in\mathcal{A}_f\left(n_{fmt}\right)$ in market $m$ of type $k$ at time $t$ is given by:

$$p_{k(m)}\left(a_{fmt}=a\,|\boldsymbol{s}_{mt},\boldsymbol{\theta}\right)=$$
$$\frac{\exp\left\{-\theta_{ec}\mathbb{1}\left\{a=1\right\}+\beta\mathbb{E}\left[V_{fkt+1}\left(\boldsymbol{s}_{mt+1},\boldsymbol{\theta}\right)\middle|\,\boldsymbol{s}_{mt},a_{fmt}=a\right]\right\}}{\sum_{a'\in\mathcal{A}_f\left(n_{fmt}\right)}\exp\left\{-\theta_{ec}\mathbb{1}\left\{a'=1\right\}+\beta\mathbb{E}\left[V_{fkt+1}\left(\boldsymbol{s}_{mt+1},\boldsymbol{\theta}\right)\middle|\,\boldsymbol{s}_{mt},a_{fmt}=a'\right]\right\}}\tag{1}$$

Our solution concept is Bayesian Nash equilibrium. In equilibrium, a bank's strategy is optimal given its beliefs about the strategies of rival banks and these beliefs are consistent with the rivals' equilibrium strategies.

# 4 Estimation

## 4.1 Market Definition

Administrative boundaries, such as Amphoes or Tambons, are unsuitable to use as a market definition in our context as they vary greatly in size. Instead, we cluster bank branch locations based on their geographic proximity to define markets. To do this, we first take the geographic coordinates of all locations

that ever had a commercial bank branch at any point in time in our data. We call these coordinates *branch locations*. We define a market cluster as a group of branch locations such that every location within the market cluster is within 5km of at least one other location in the same cluster. For example, if a single branch location is more than 5km from every other branch location in the country, then that location is in a cluster by itself. If two branch locations are within 5km of each other but neither of the two are within 5km of any other location in the country, those two locations form a single market cluster. If three branch locations were in a straight line, each 4km from each other, then all three would form a single market cluster, even though the two branches on either end are 8km away from each other.

To construct the market clusters in practice, we construct an $L \times L$ boolean matrix where element $(\ell, \ell')$ equals one if branch locations $\ell$ and $\ell'$ are within 5km of each other and is zero otherwise. We multiply this boolean matrix by itself until it stops changing. The $\ell$th row of this matrix gives the locations in the same market as location $\ell$.

Figure 5 shows an example of our clustering approach in the south of Thailand. Points within the same diamond that are the same color are clustered in the same market. There are a large number of markets with only one or two locations, but also some markets with many locations.

Out of the 4,128 commercial branches that were ever active in our data, this approach generates 731 markets. As our data do not include the greater Bangkok area, we omit 11 markets where there was at least one branch locations within 5km of the border of either the Bangkok Metropolitan or Samut

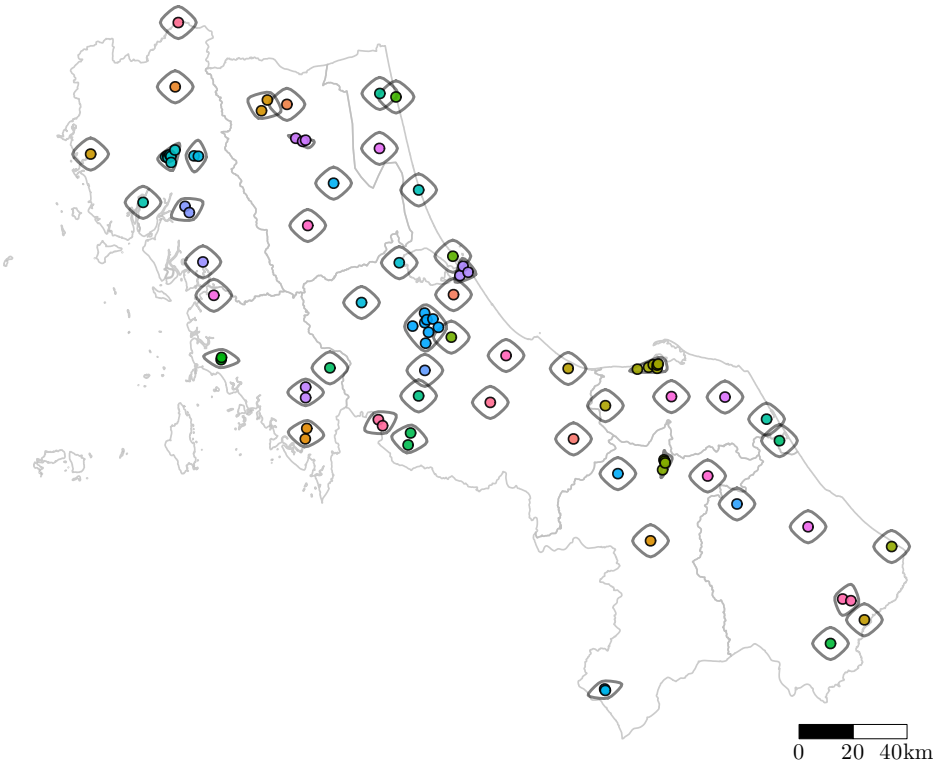



FIGURE 5: Clustering Locations Example in Southern Thailand

Prakan provinces. The size of the state space in our model increases exponentially with the maximum number of possible branches a single bank can have in a market. We therefore set the maximum number of branches $N$ to 3 and omit 43 markets where one of the four largest banks had more than 3 active branches at any point in time. Finally, to ensure the branch networking strategies coincide with the strategy space in the model, we omit 2 additional markets where one of the banks opened more than one branch in a single year. We estimate our model with the 675 remaining markets. The locations of all the markets we use in estimation are shown in Figure 6.

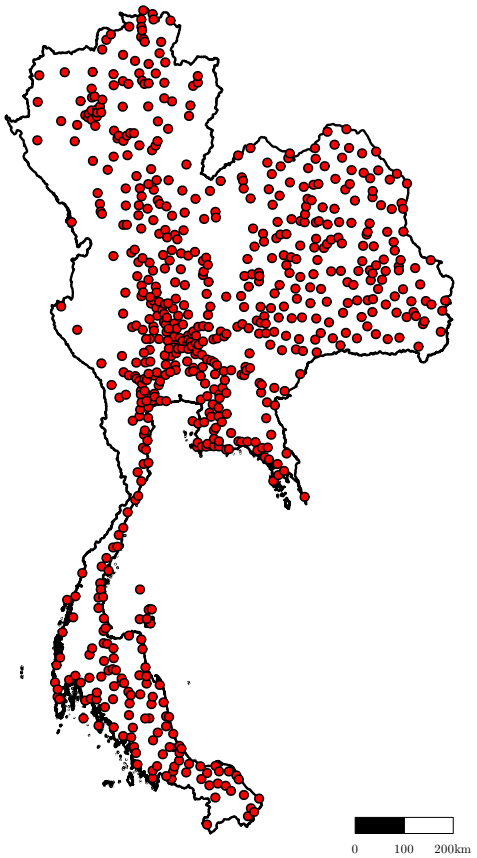


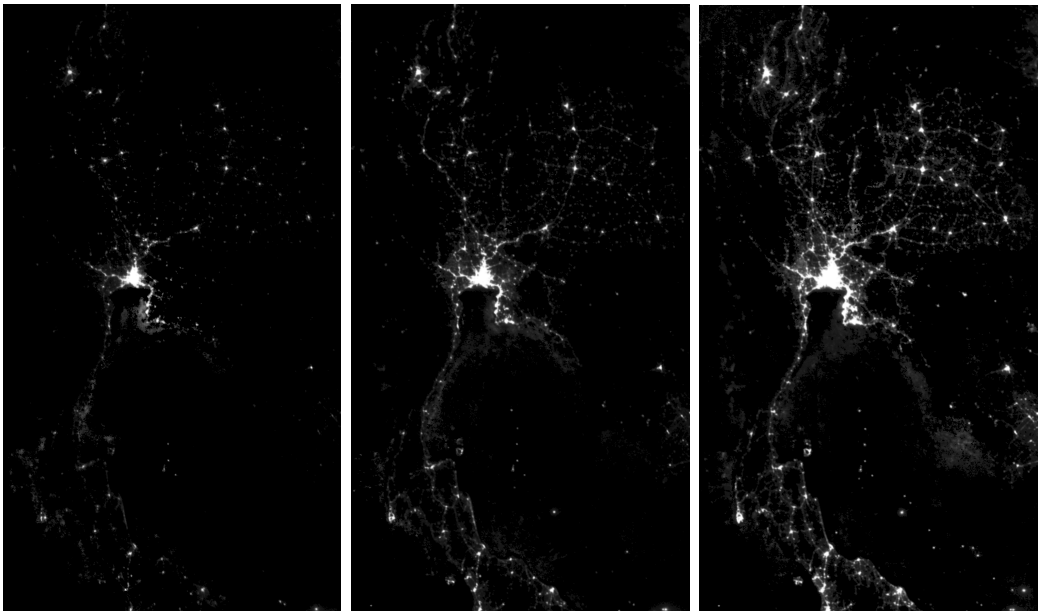

FIGURE 6: Centroid of Market Locations used in Estimation.

## 4.2 Market Demand Shifter: Nighttime Luminosity

Accurate proxies for demand are not readily available at a fine geographic level for Thailand. We instead use nighttime luminosity data from the National Oceanic and Atmosphere Administration to proxy market attractiveness. These data have been used as proxies for population and income in a large number of applications (see for example, Henderson et al. (2011, 2012), Bleakley and Lin (2012), Gennaioli et al. (2013), Michalopoulos and Papaioan-

nou (2013)). These data come from satellite images captured by the US Air Force at night between 8:30 PM and 10:00 PM local time around the world. These images are then processed and cleaned to represent the average amount of light emanating from a geographic location during a year. Observations obstructed by clouds are excluded, as well as observations with light coming from forest fires, gas flares, sunlight (from the summer months) and moonlight. Values are represented on a scale that ranges from 0 to 63 that measures the amount of light captured by the camera's sensor. This scale is bottom- and top-coded, with very rural locations being bottom-coded at 0 and dense urban areas being top-coded at 63. Top-coding is not a large issue in Thailand, with only 0.27% of the country being top-coded in the final year of data. Furthermore, our analysis focuses on rural areas where there is no top-coding. Data are available from 1992-2013 and are represented on a grid with a 30 arc-second resolution. In Thailand, one cell of the nighttime luminosity data is at a resolution of approximately 900m$\times$900m. Because our bank branch data are available until 2010, we constrain our sample period in estimation to 1992-2010, the overlap of the two data sets.

Figures 7a to 7c show the nighttime luminosity in Thailand in the first, middle and last year of our sample period. The brightest area in the center is Bangkok. The bright lights south of Bangkok in the Gulf of Thailand are not measurement error; rather they are from squid fishing boats that shine bright green LED lights to attract plankton to the surface. As these observations are in the sea, they are not counted in our measurement of demand at branch locations.



(A) 1992 (B) 2001 (C) 2010

FIGURE 7: Night lights data over time.

Because our structural model uses temporal variation in nighttime luminosity within markets, it is necessary to first intercalibrate the digital number values across years (Wu et al., 2013). When satellites are replaced, luminosity values may change even if there is no change in luminosity. We intercalibrate the nighttime luminosity values as follows. Let $Y_t$ be Thailand's aggregate GDP PPP (in US$) in year $t$ and let $NL_t$ be the the sum of nighttime luminosity values within the country's borders in year $t$. When two satellite readings are available in a year, $NL_t$ is the average of the two satellites. The multiplier for year $t$ is then calculated as:

$$m_t = \left( \frac{\sum_{s=1992}^{2013} Y_s}{\sum_{s=1992}^{2013} NL_s} \right)^{-1} \frac{Y_t}{NL_t}$$

The multiplier ensures that aggregate nighttime luminosity follows the same trend as aggregate GDP and is scaled such that the sum of the intercalibrated nighttime luminosity values matches the sum of the raw values.

We calculate the demand shifter measure for our markets by drawing a circle with a radius of 10km around the centroid of branch locations within a market and summing the values of the nighttime luminosity digital numbers within that circle. More specifically, let $(x_m, y_m)$ be the longitude and latitude of the centroid of branch locations in market $m$ and let $d\left((x,y),(x_m,y_m)\right)$ be the Haversine distance in kilometers between the pairs of coordinates $(x,y)$ and $(x_m,y_m)$. The nighttime luminosity value for a particular market $m$ at time $t$ is then:

$$nl_{mt} = m_t \int_{-90}^{90} \int_{-180}^{180} \mathbb{1}\left\{d\left((x,y),(x_m,y_m)\right) \leq 10\right\} nl\left(x,y\right) dxdy$$

where $nl\left(x,y\right)$ is the nighttime luminosity digital number at point $(x,y)$. We set nighttime luminosity values outside of Thailand's borders to zero before performing these calculations to avoid including the large values from the squid-fishing boats.

This calculation is illustrated in Figure 8. The market shown has four branch locations illustrated with four red circles. Three of the branches are located close together, whereas one of the branches is located approximately 4km farther away to the south-west. The green circle has a radius 10km around the centroid of the market. The demand shifter is the sum of the nighttime luminosity digital numbers in the entire circle.

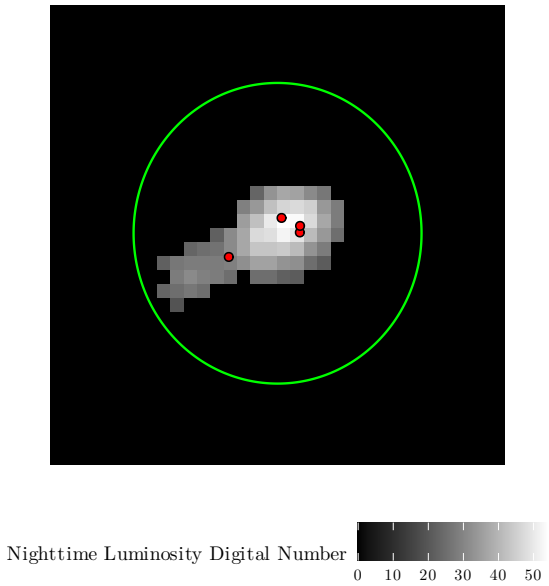



FIGURE 8: Night lights within a 10km radius of market centroid.

To evaluate how well our market demand shifter measures local GDP, we obtain the provincial GDP from Thailand's Office of the National Economic and Social Development Council. The province (changwat) is the smallest geographic unit where local GDP values are available. We compare provincial GDP values with the sum of intercalibrated nighttime luminosity values within a province. The two variables are strongly related and have a correlation of 0.78. A scatter plot of the two variables is shown in the Online Appendix.

In the structural model, all branches entering in a market have the same value of the demand shifter. In our modeling, we have experimented with allowing banks to open branches in specific locations within the market cluster and allowed the value of demand to differ by location within a market. We did this by summing the values of nighttime luminosity in a radius around

each branch location rather than around the market centroid. We found that the values of demand were highly correlated across locations within market clusters in a year. The assumption that all branches in the same market experience the same value of the demand shifter therefore greatly reduces the computational complexity of the model, without sacrificing substantial within-market variation in demand.

## 4.3 Market Types

We classify each market into one of five types. We do this by using the estimated market fixed effects from an ordered probit regression, similar to how Igami and Yang (2016) obtain starting values for their model. The ordered probit regression is at the bank-market-year level. The dependent variable is the bank's action $a_{fmt}$, which takes on the values $-1$, 0 or 1 depending on if bank $f$ closed, did nothing or opened a branch in market $m$ in year $t$. For explanatory variables we include our nighttime luminosity market demand shifter, the number of the bank's own branches, the number rival branches, and a market fixed effect.

Table 1 shows the coefficient estimates from this regression. The regression shows that banks are more likely to open branches when local demand is greater. They are less likely to enter in the presence of their own branches and branches of rival banks. Figure 9 shows a histogram of the estimated market fixed effects from this regression. We divide markets into five equally-sized categories based on the value of the estimated market fixed effects, with market type 1 having the smallest values.

| *Dependent variable:* | Enter/Nothing/Exit | |
|---|---|---|
| Nighttime luminosity | 0.230 | (0.016) |
| Own branches | −1.165 | (0.052) |
| Rival branches | −0.294 | (0.028) |
| Number of observations | 54080 | |
| Market fixed effects | Yes | |

Standard errors in parentheses.

TABLE 1: Ordered Probit regression results.

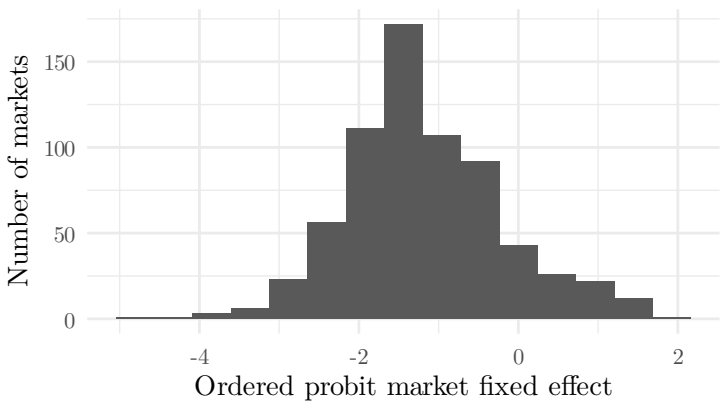



FIGURE 9: Market fixed effects

Higher market types have more branches on average, but they are not necessarily more populated or have higher local GDP. In fact, the average nighttime luminosity values for type 1 markets (the lowest type) is more than twice than the average among other types. We use the market types to capture persistent unobserved heterogeneity between markets that is not captured by nighttime luminosity.

In bank's profit function, we assume the presence of branches of the two large government banks (the BAAC and GSB) do not affect the commercial banks' profits. This is because these banks mainly serve different sectors of the market. Assuncao et al. (2020) also show that the BAAC's branching

| *Dependent variable:* | Enter/Nothing/Exit | | | |
|---|---|---|---|---|
| Nighttime luminosity | 0.153 | 0.152 | 0.153 | 0.152 |
| | (0.007) | (0.007) | (0.007) | (0.007) |
| Own branches | −0.865 | −0.865 | −0.865 | −0.865 |
| | (0.042) | (0.042) | (0.042) | (0.042) |
| Rival branches | −0.038 | −0.037 | −0.037 | −0.037 |
| | (0.019) | (0.019) | (0.019) | (0.019) |
| Crisis | −0.599 | −0.599 | −0.599 | −0.599 |
| | (0.046) | (0.046) | (0.046) | (0.046) |
| BAAC presence | | −0.056 | | −0.057 |
| | | (0.046) | | (0.048) |
| GSB presence | | | −0.014 | 0.001 |
| | | | (0.041) | (0.043) |
| Market type effects | Yes | Yes | Yes | Yes |
| Bank fixed effects | Yes | Yes | Yes | Yes |
| Number of observations | 54080 | 54080 | 54080 | 54080 |

Standard errors in parentheses.

TABLE 2: Ordered Probit regression results with market types.

decisions are not consistent with profit maximization. We argue that our market types capture the presence of these branches and their inclusion to the banks' profit functions would not affect our main results. To test for this, we estimate an ordered probit regression of the branching problem, similar to that in Table 1, where we include all the variables in the bank's profit function. The results are shown in the first column of Table 2. When we add the presence of BAAC or GSB branches, the coefficients of the other parameters remain virtually identical and the coefficients on BAAC and GSB presence are not statistically significant. Therefore we omit the BAAC and GSB branches from our structural model, which greatly reduces the size of our state space.

## 4.4 Structural Parameter Estimation

We now discuss how we estimate our vector of structural parameters:

$$\boldsymbol{\theta} = \left(\{\theta_k\}_{k=1}^{k=5}, \{\theta_{bf}\}_{f=2}^{f=4}, \theta_z, \theta_{own}, \theta_{comp}, \theta_{crisis}\right)$$

We do not estimate the annual discount factor but set its value to $\beta = 0.95$ ex-ante. This discount factor is commonly used in the literature (for example, Holmes (2011), Dunne et al. (2013), Collard-Wexler (2013) and Zheng (2016)).

The state space is $\boldsymbol{s}_{mt} = \left(\{n_{fmt}\}_{f=1}^{F}, z_{mt}, \zeta_t\right)$. The transition probabilities of $n_{fmt}$ are endogenously determined within our model, but we specify the transitions of the crisis indicator, $\zeta_t$, and the market demand shifter, $z_{mt}$, ex-ante. In the period following the crisis, branch profits fall by $\theta_{crisis}\zeta_t$. We do not assume that banks form expectations over the evolution of $\zeta_t$. Rather, we assume that in the years before the crisis, banks expect $\zeta_t = 0$ in all future time periods. Once the crisis arrives, $\zeta_t$ is deterministic and banks have full knowledge of its process. During the crisis, $\zeta_t = 1$ for seven time periods. We have found this choice produces aggregate branching patterns that best match the patterns in the data. After the crisis, banks do not expect there will be another large crisis and $\zeta_t = 0$ in all future time periods.

Banks place positive probability on drops in the market demand shifter, $z_{mt}$, in all time periods. We discretize the sum of nighttime luminosity values within a 10km circle around a market centroid, $z_{mt}$, into 10 bins, where each bin has the same number of observations in the data. We assume the transition probabilities for nightlights $p_{zt}(z'|z)$ vary over time, but are the same across

all market types. We use a simple frequency estimator to estimate $p_{zt}(z'|z)$:

$$p_{zt}(z'|z) = \frac{\sum_{m=1}^{M} \mathbb{1}\{z_{mt} = z, z_{mt+1} = z'\}}{\sum_{m=1}^{M} \mathbb{1}\{z_{mt} = z\}}$$

It is possible that the opening of bank branches could affect the level of nighttime luminosity within a market directly, and so these transition probabilities may depend on the banks' branching decisions. The improved access to credit from a new branch could lead to the opening of new businesses which would increase the amount of nighttime light emanating from a market. However, we argue that the effect of branch entry on these transition probabilities would be small and assume banks do not take into account the effect of their actions on the transition probabilities. Young (2019) uses a regression discontinuity design in India to study the effect of bank branch expansion on agriculture, manufacturing and local GDP. He finds that treated districts, which had an average increase of 10 branches, had a 1.65% increase in nighttime luminosity over a 6-year period. Based on these findings, the effect of a single additional branch would only have a very small effect on the year-to-year transition probabilities.

Given a particular trial value of the structural parameters, we solve the model by backward induction. We assume the terminal time period $T$ is 20 periods in the future. Starting with the terminal period and working backwards, we solve for the equilibrium choice probabilities within each time period for each market type. Like Augereau et al. (2006), we do not restrict strategies to be symmetric across banks. However, the same bank is restricted to have

the same strategy function in all markets of the same type.

We use maximum likelihood to estimate the structural parameters. Let $a_{fmt} \in \{-1, 0, 1\}$ be the action chosen by firm $f$ in market $m$ of type $k$ at time $t$ in the data, where the sample period is 1992 to 2010 ($\widetilde{T} = 19$). The maximum likelihood estimator of $\boldsymbol{\theta}$ is then:

$$\widehat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} \sum_{t=1}^{\widetilde{T}} \sum_{m=1}^{M} \sum_{f=1}^{F} \log \left( p_{k(m)} \left( a_{fmt} \left| \boldsymbol{s}_{mt}, \boldsymbol{\theta} \right. \right) \right)$$

where $p_{k(m)} \left( a_{fmt} | \boldsymbol{s}_{mt}, \boldsymbol{\theta} \right)$ is the equilibrium conditional choice probability for market $m$'s market type in state $\boldsymbol{s}_{mt}$ given parameters $\boldsymbol{\theta}$.

# 5 Model Estimates

Table 3 shows the structural parameter estimates. The estimates show a similar pattern to the reduced-form ordered probit regression in Table 2. Branch profits are increasing in nighttime luminosity and decreasing in the presence of own and rival branches. The estimated effect of the crisis shows a large decrease in profits, much greater than the presence of rival branches. The estimated constant is monotonically increasing in the market type, where the higher market type had the higher ordered probit market fixed effects. The estimates of the bank-specific profit shifters $\theta_f$ are close to zero except for Siam Commercial Bank. This estimate is negative relative to the base bank of Krung Thai because this is the smallest of the four largest banks.

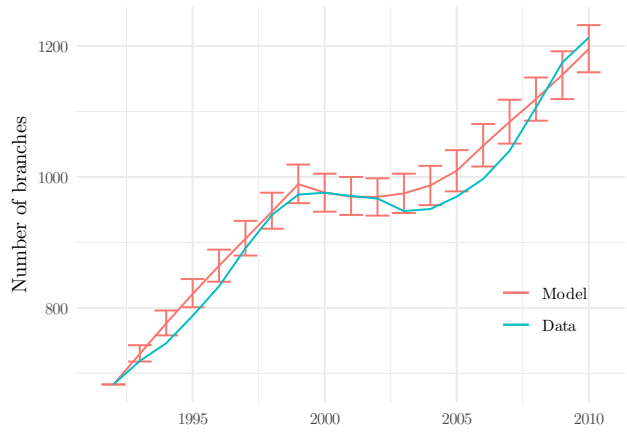
We can use the banks' annual reports to interpret the magnitudes of the estimated parameters. The average profits per branch from our four banks in

| | Estimate | Standard Error |
|---|---|---|
| Entry cost | 11.279 | (0.214) |
| Market type 1 | 0.098 | (0.020) |
| Market type 2 | 0.221 | (0.015) |
| Market type 3 | 0.284 | (0.014) |
| Market type 4 | 0.361 | (0.014) |
| Market type 5 | 0.459 | (0.017) |
| Bank 2 | 0.001 | (0.008) |
| Bank 3 | 0.003 | (0.008) |
| Bank 4 | −0.057 | (0.009) |
| Nighttime luminosity | 0.049 | (0.003) |
| Own branches | −0.064 | (0.005) |
| Rival branches | −0.015 | (0.004) |
| Crisis | −0.565 | (0.044) |

Table 3: Structural Parameter Estimates.

2010 was US$701,230. Using the average profits of active branches in 2010 according to our model, one unit in the parameter estimates is approximately US$1.57m. Based on this value, the presence of one more rival branch on average lowers profits by US$26,700 per year.

To show how our model fits with the data, we solve for the equilibrium strategies at the estimated structural parameters and simulate branch network expansion paths based on these strategies. Figure 10 shows the average total number of active branches from 1,000 of such simulations. The error bars represent the 0.025 and 0.975 quantiles of the simulated network expansion paths. We can see that the predicted total number of branches match the aggregate temporal patterns in the data relatively well. In the Online Appendix, we show the aggregate number of branches split by market type. Only in market type 5 do the predicted entry paths slightly overpredict the number of branches in certain years. In the Online Appendix, we also show




Error bands contain 95% of simulated paths from 1,000 simulations.

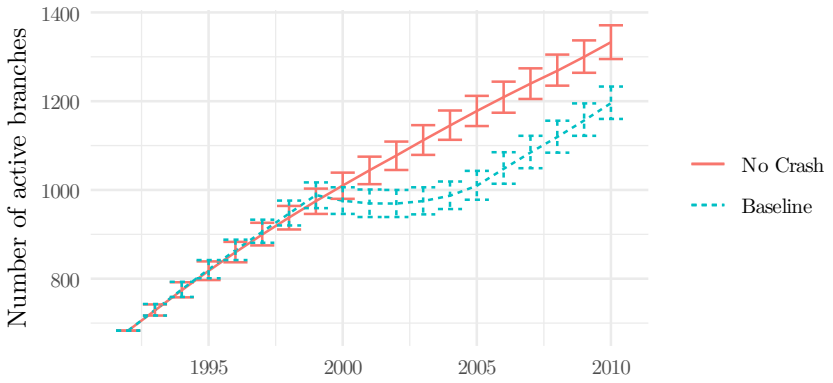
FIGURE 10: Predicted number of active branches versus data by market type.

how the total number of branch openings and closings per year predicted by the model compare with the data. In general, the model captures total branch openings and closings well. However, the model predicts the peak of branch closings to occur in 1999, whereas in the data the peak occurs in 2001.

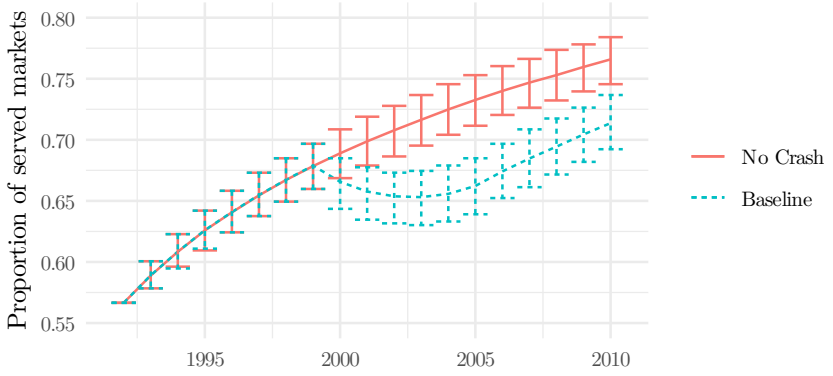
# 6 Counterfactual Simulations

## 6.1 No Financial Crisis

We now use this model to simulate counterfactual experiments to understand the effect of the financial crisis on banks' branching strategies. For our first counterfactual experiment, we simulate the expansion of the bank branch network if the financial crisis of 1997 did not occur. We first set the nighttime luminosity values for each market in 1997-2001 to their values in 1996 and ad-




(A) Number of active branches.




(B) Proportion of served markets.

FIGURE 11: Branch network expansion under no crash versus baseline.

just the transition probabilities accordingly.[1] We also set our crash indicator $\zeta_t$ equal to zero for all time periods. In our counterfactual, the crash does not occur and firms also do not place a positive probability of it occurring in the future. We then solve for the equilibrium strategies of the banks by backward induction and simulate network expansion paths using these strategies.

[1]The average of the counterfactual nighttime luminosity paths is shown in the Online Appendix. If we were to instead assume that nighttime luminosity continued according to its pre-crash trend, the counterfactual paths would be much higher and the effect of the crisis would be even more severe. Therefore the assumption we make is more conservative.

Figure 11 shows the results from 1,000 simulations according to these equilibrium strategies. Figure 11a shows the average number of branches on aggregate from our simulations, together with error bands that contain 95% of the simulations. The baseline model predictions are also shown for comparison purposes. We can see that the number of active branches continued according to the pre-crash trend in the absence of the crisis. By 2007, ten years after the crisis, there were 14.3% more branches.

We are not only interested in the total number of branches, but also the proportion of markets served by at least one branch, as markets without any branches have poorer access to credit. For each of our simulated network expansion paths, we also calculate the proportion of markets that had at least one branch from the banks in our model. We do this under the no-crash counterfactual and under the estimated model parameters. This is shown in Figure 11b, together with error bars that contain 95% of our simulations. We can see that in the years following a crash, the number of markets served fell and did not recover until 2007. However, under the no-crash counterfactual, the proportion of served markets continued according to the pre-crash trend, with 9.1% more markets served by 2007 compared to the baseline scenario.

Markets which saw their branches close may still have access to branches in nearby markets. We calculate the distance to the nearest market with a branch under this counterfactual scenario and find that the crash caused the average distance to increase by 23.6% ten years after the crisis.[2] Ji et al. (2021) estimate a regression model using Thai data explaining the percentage

[2]Although we exclude a subset of markets in estimation, we use the full set of markets to perform this calculation.

of village residents with access to commercial loans by the log of travel distance. Using their estimated effect with our predicted change in distance, village access to commercial loans would have been higher by 10.5 percentage points in the absence of the crisis.

We also compare our model's predictions under an alternative assumption of banks' expectations of the evolution of the crisis indicator. Instead of assuming that banks know when the crisis period will end, banks believe it will last forever. Once the crisis arrives, banks believe that $\zeta_t = 1$ in all future time periods. Once the crisis is over, banks believe that $\zeta_t = 0$ in all future time periods. We estimate the parameters of the model according to this assumption and run the same counterfactual experiment as above. We find the effect of the crisis to be more severe under this alternative assumption. We find there are 16.9% more branches and 10.5% more served markets after 10 years. The average distance to the nearest branch increases by 25.8% and the predicted change in financial access is 11.6 percentage points. Further details of these results are in the Online Appendix.[3]

## 6.2 Bank Branch Supports

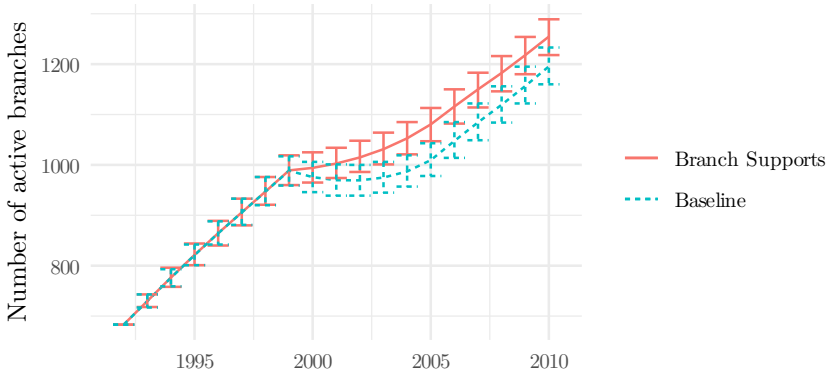
For our second counterfactual experiment, we consider the effect of bank branch supports. Banks that choose to close a branch in a location during the crisis period may never reopen a branch in that location, even if the

[3]We also compare our model's predictions under the assumption that the four banks behave as a cartel. If we re-estimate the model's parameters under this assumption, ten years after the crisis we would observe 12.7% more branches, 9.2% served markets and an increase in financial access of of 10.5 percentage points had the crash not occurred. The results are shown in the Online Appendix.

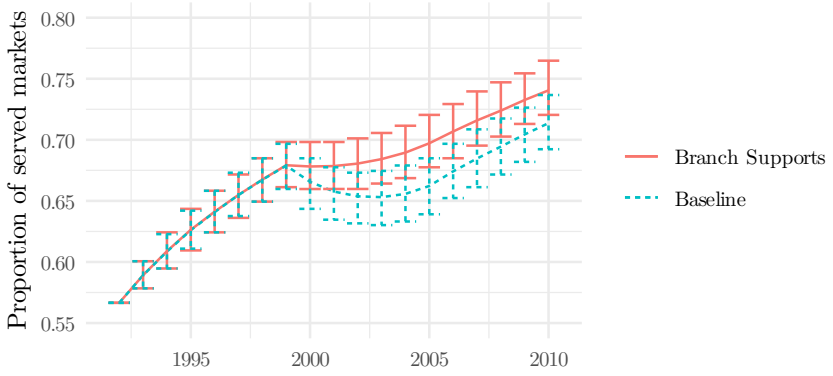
branch they closed would have been profitable after the crisis was over. This is because our estimated entry cost is on average 25 times the annual profits for a branch. Banks may also have closed branches during the crisis that were long-term profitable because of liquidity issues. If branches were supported with subsidies for the duration of the crisis period, markets that saw their branches close may instead continue to have a branch during and also after the crisis period.

We set the branch support subsidy equal to half of the estimated losses from our post-crisis indicator in the branch's profit function. Specifically, each active branch's profits are increased by $\frac{1}{2}\theta_{crisis}\zeta_t$ each period during the crisis. According to our estimated value for each unit in model's profit function using the banks' annual reports, this subsidy amounts to US$443,394 per branch annually over five years. We retain the same values for nighttime luminosity as in the data for this counterfactual. The crisis indicator also captures the effect of lower liquidity on the banks' branching strategies. Therefore this counterfactual can also be interpreted as easing the liquidity issues faced by the banks.

The results are shown in Figure 12, presented in the same format as Figure 11 for ease of comparison. Figure 12a shows that although the total number of branches did not continue according to its pre-crash trend, the total size of the branch network did not decrease following the crisis. Ten years following the crisis, the total number of branches is approximately 6.0% higher. Similarly, Figure 12b shows that the subsidy prevented the proportion of served markets from decreasing, but with fewer markets served compared to the no




(A) Number of active branches.




(B) Proportion of served markets.

FIGURE 12: Branch network expansion under bank branch support subsidy versus baseline.

crash scenario. By 2007, the proportion of served markets was 4.6% higher. These results show that targeting particularly vulnerable markets with branch supports could prevent these locations from being unbanked after the overall economy recovers from the crisis.

# 7 Conclusion

In this paper, we argue that the effect of financial crises on bank branch location choices provides an unexplored channel by which crises affect access to credit. Because opening new branches entails a large up-front investment, markets that see branches close during the crisis may go unbanked for many years after the overall economy recovers. We study this issue in the context of the 1997 Thai financial crisis by estimating a dynamic structural model of banks' branching strategies. In the model, we allow for complementarity in payoffs for bank branches in nearby locations, as well as competitive effects between rival banks. Using this model we predict the evolution of bank branch locations under the counterfactual scenarios of no financial crisis in 1997 and with bank branch support subsidies. We find that the financial crisis had large impacts on the total number of branches and the proportion of markets served by at least one branch. We find that there would have been 14.3% more branches and 9.1% more markets with at least one branch after ten years had the crisis not occurred. Per-branch subsidies equal to approximately half of an average branch's profits could have prevented the proportion of markets served by a branch from falling below pre-crisis levels.

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# A Online Appendix

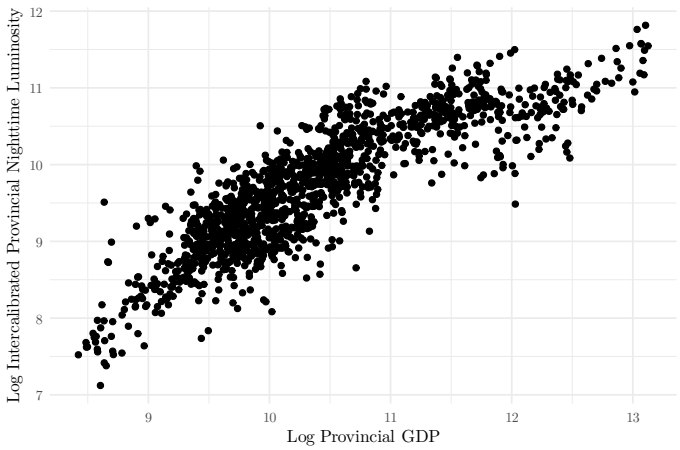
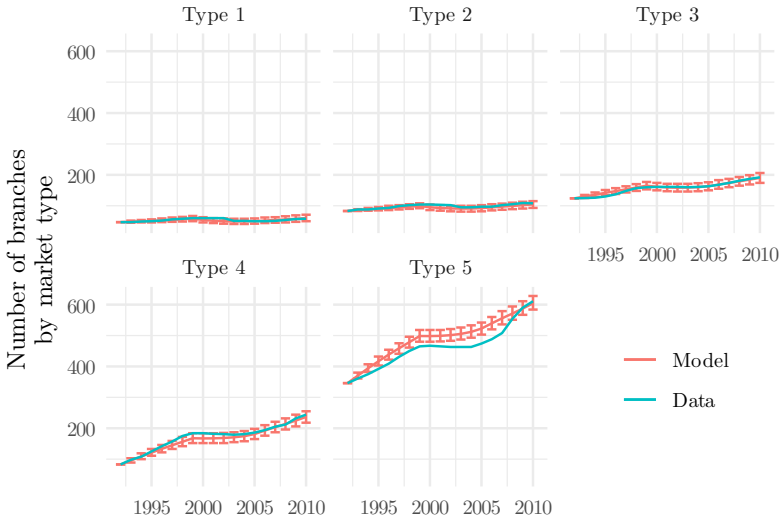
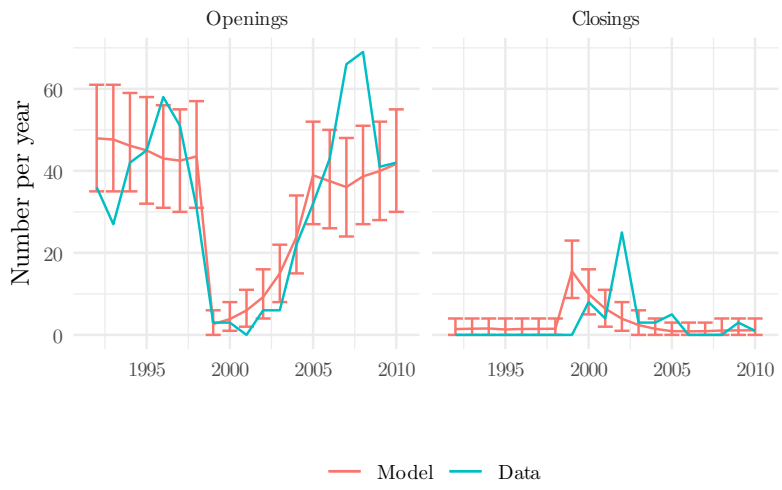



FIGURE A.1: Log provincial GDP versus log provincial nighttime luminosity.




Error bands contain 95% of simulated paths from 1,000 simulations.

FIGURE A.2: Predicted number of active branches versus data by market type.




Error bands contain 95% of simulated paths from 1,000 simulations.

FIGURE A.3: Predicted number of openings and closings per year versus data.

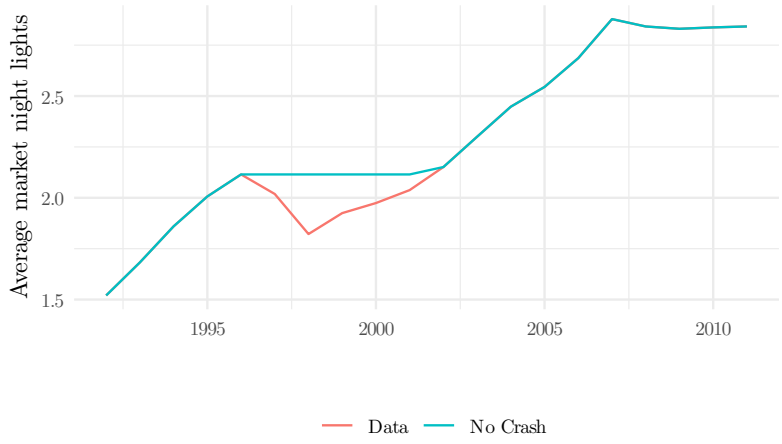
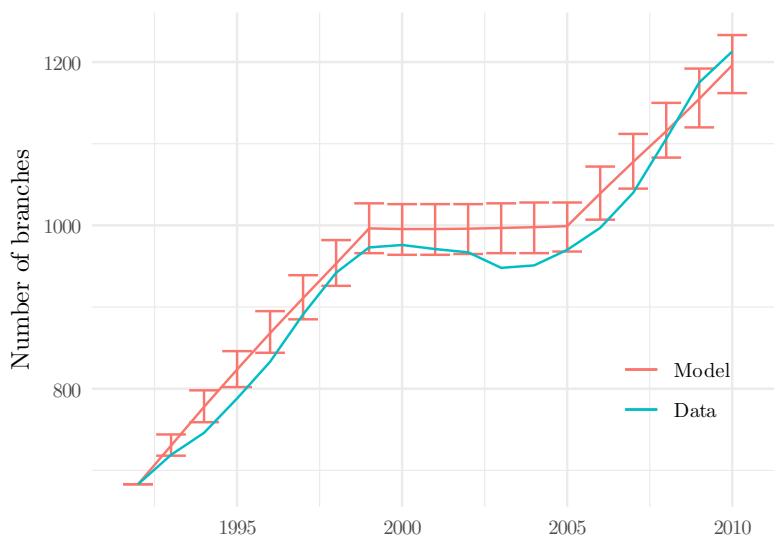



FIGURE A.4: Average nighttime luminosity across markets in the data and under the counterfactual scenario of no crisis.

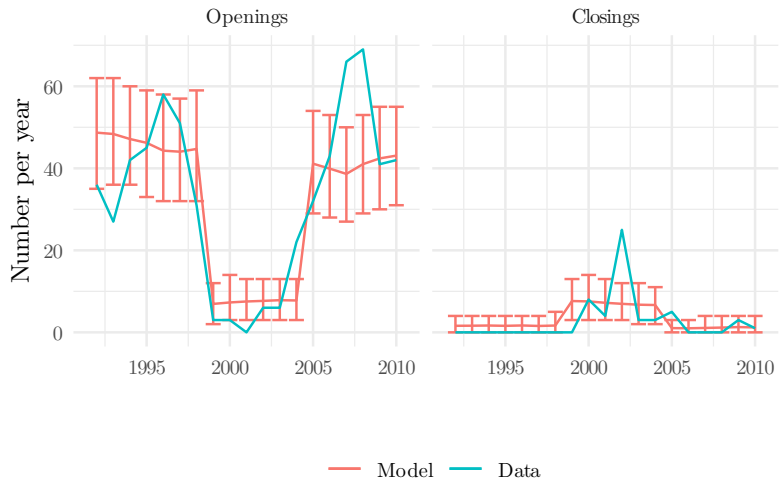
| | Estimate | Standard Error |
|---|---|---|
| Entry cost | 11.058 | (0.200) |
| Market type 1 | 0.090 | (0.020) |
| Market type 2 | 0.230 | (0.014) |
| Market type 3 | 0.282 | (0.013) |
| Market type 4 | 0.352 | (0.013) |
| Market type 5 | 0.437 | (0.016) |
| Bank 2 | 0.001 | (0.008) |
| Bank 3 | 0.007 | (0.008) |
| Bank 4 | −0.058 | (0.009) |
| Nighttime luminosity | 0.046 | (0.003) |
| Own branches | −0.060 | (0.004) |
| Rival branches | −0.011 | (0.004) |
| Crisis | −0.129 | (0.009) |

TABLE A.1: Structural estimates when assuming banks believe the crisis will last forever.




Error bands contain 95% of simulated paths from 1,000 simulations.

FIGURE A.5: Number of active branches under the assumption that banks believe the crisis will last forever.



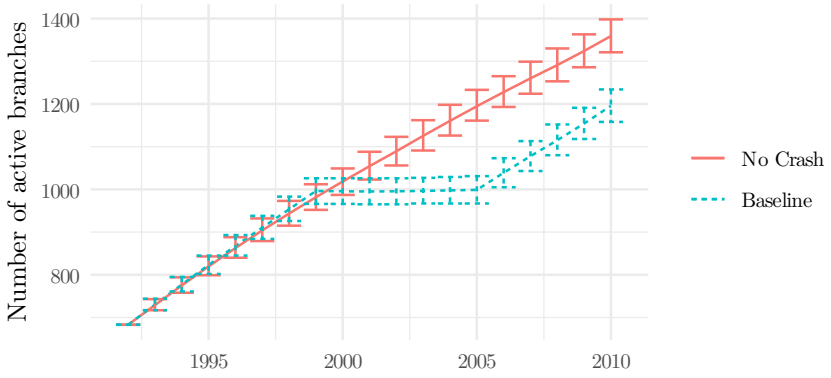

Error bands contain 95% of simulated paths from 1,000 simulations.

FIGURE A.6: Number of openings and closings under the assumption that banks believe the crisis will last forever.

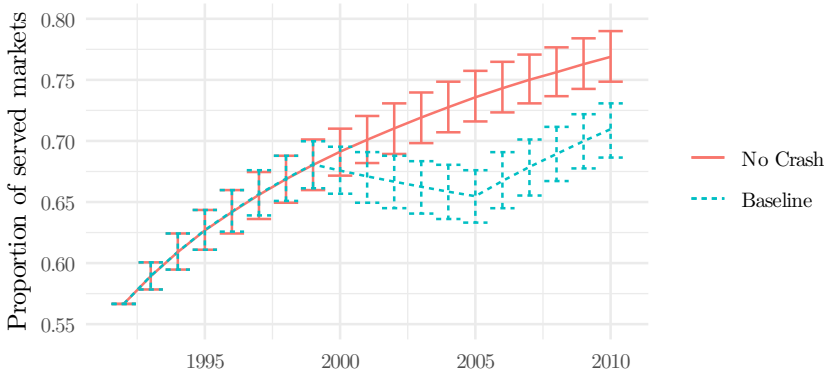
| | Estimate | Standard Error |
|---|---|---|
| Entry cost | 9.633 | (0.232) |
| Market type 1 | 0.045 | (0.024) |
| Market type 2 | 0.197 | (0.016) |
| Market type 3 | 0.277 | (0.015) |
| Market type 4 | 0.374 | (0.015) |
| Market type 5 | 0.514 | (0.022) |
| Nighttime luminosity | 0.064 | (0.005) |
| Own branches | −0.035 | (0.003) |
| Crisis | −0.584 | (0.050) |

To estimate the model's parameters under the assumption that the banks behave as a cartel, we sum the number of branches of each bank and assume a single bank behaves as a monopolist. We do not estimate $\theta_{comp}$ in this specification because the monopolist only competes with itself.

TABLE A.2: Parameter estimates assuming that banks behave as a cartel.

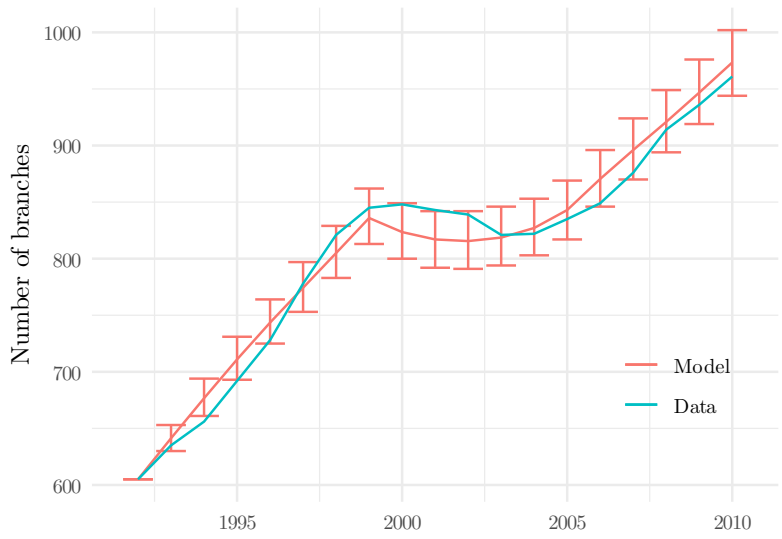



(A) Number of active branches.



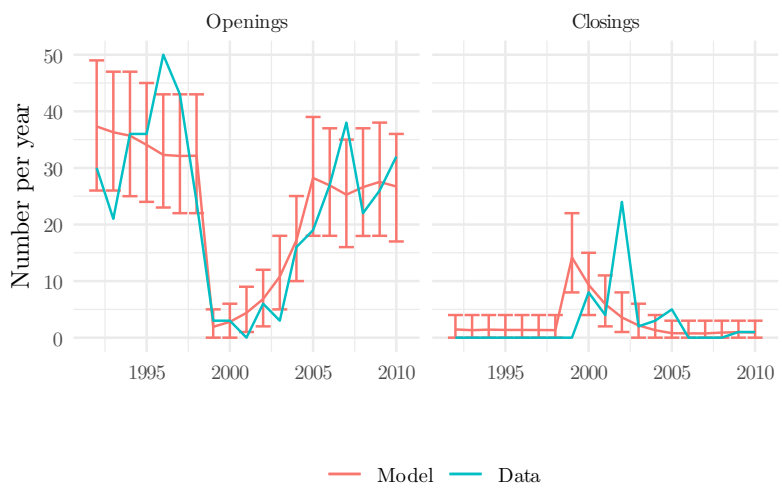

(B) Proportion of served markets.

FIGURE A.7: Branch network expansion under no crash versus baseline under the assumption that banks believe the crisis will last forever.



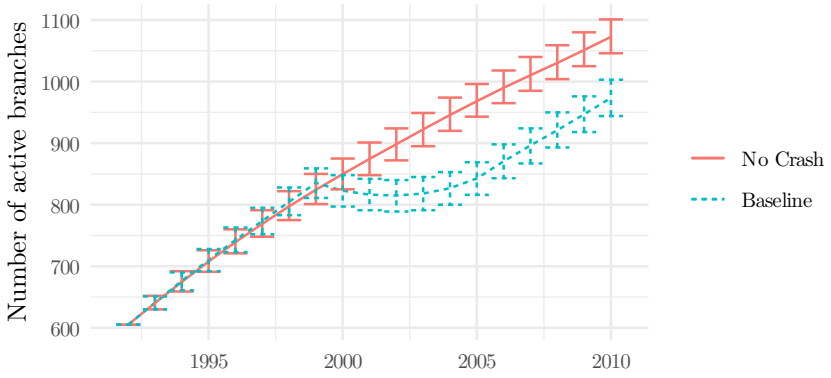

Error bands contain 95% of simulated paths from 1,000 simulations.

FIGURE A.8: Number of active branches under the assumption that banks behave as a cartel.

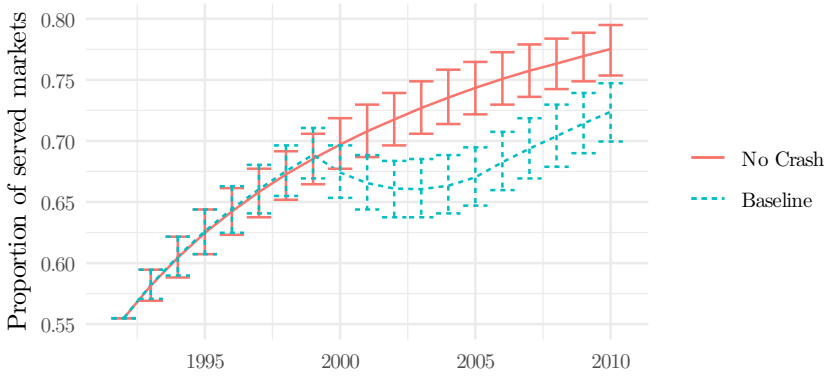



Error bands contain 95% of simulated paths from 1,000 simulations.

FIGURE A.9: Number of openings and closings under the assumption that banks behave as a cartel.




(A) Number of active branches.




(B) Proportion of served markets.

FIGURE A.10: Branch network expansion under no crash versus baseline under the assumption that banks behave as a cartel.